RISHI THAKUR

DATA SCIENTIST

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PROFESSIONAL SUMMARY

Data Scientist with a strong quantitative foundation (M.Tech CSE, GATE CS qualified) and expertise in exploratory data analysis (EDA), statistical inference, machine learning, and predictive modeling. Proven track record handling severe class imbalances via SMOTE + Tomek Links and deploying predictive regression and classification pipelines. Adept at translating operational bottlenecks into rigorous statistical models and executive dashboards.

TECHNICAL SKILLS

- **Machine Learning & Statistics:** Machine Learning (Random Forest, XGBoost, KMeans, Regression), Feature Engineering, Resampling (SMOTE + Tomek Links), Hypothesis Testing, R^2, RMSE, F1-Score
- **GenAI, Deep Learning & AI Tools:** Claude Code, Google Antigravity, Prompt Engineering, Agentic Workflows, Diffusion Models (DDPM/DDIM), Transformers, BERT Embeddings, PyTorch, TensorFlow
- **Programming & Data Systems:** Python (Pandas, NumPy, Scikit-learn), Advanced SQL (Joins, CTEs, Window Functions), MySQL, MariaDB, OracleDB, Microsoft Power BI (DAX), Linux/Bash, Git

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
Data Science & Analytics Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Conducted exploratory data analysis (EDA) across **33 districts** on healthcare attendance records, addressing schema anomalies and duplicate designations.
- Engineered an automated **SQL/ETL feature pipeline** mapping AEBAS biometric IDs to State Treasury payroll codes, establishing cross-departmental ground-truth records.
- Formulated a **3-tier risk classification model** tracking bonded medical personnel across facilities, pinpointing critical compliance gaps.
- Reconciled disparate database schemas across MariaDB, State Treasury, and e-Hospital systems with **100% data fidelity**.
- Architected an executive 6-page **Power BI dashboard** featuring **4-tier drill-down hierarchies** (District → Facility → Designation → Individual) to present KPIs to leadership.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Predictive Modeling & Telemetry*
Technologies: Python, PyTorch, GRU, DDPM/DDIM, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Architected a predictive engine pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** simulating 50 paths to forecast signal degradation from multivariate **spatio-temporal** telemetry.
- Resolved extreme **98%/2% class imbalance** using a hybrid **SMOTE + Tomek Links** pipeline, implementing a two-stage Random Forest classifier to attain a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Applied NLP & Deep Learning*
Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, TensorFlow, Scikit-learn

- Engineered dual NLP classification architectures: extracted 768-dim dense contextual embeddings from pre-trained **BERT** paired with **XGBoost**, and constructed a deep hybrid Bi-LSTM/Transformer network.
- Applied stratified sampling (80/20), text normalization (NFKD), dropout (0.5), and dynamic learning rate decay, achieving **99% Precision** and **98% Recall** in synthetic LLM classification.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **Celonis Process Mining Foundations** (Credly)
- **Microsoft Power BI for Business Intelligence** (Udemy)
- **SQL for Beginners & Database Design** (Udemy)
- **IELTS Academic:** 7.5 Band Score